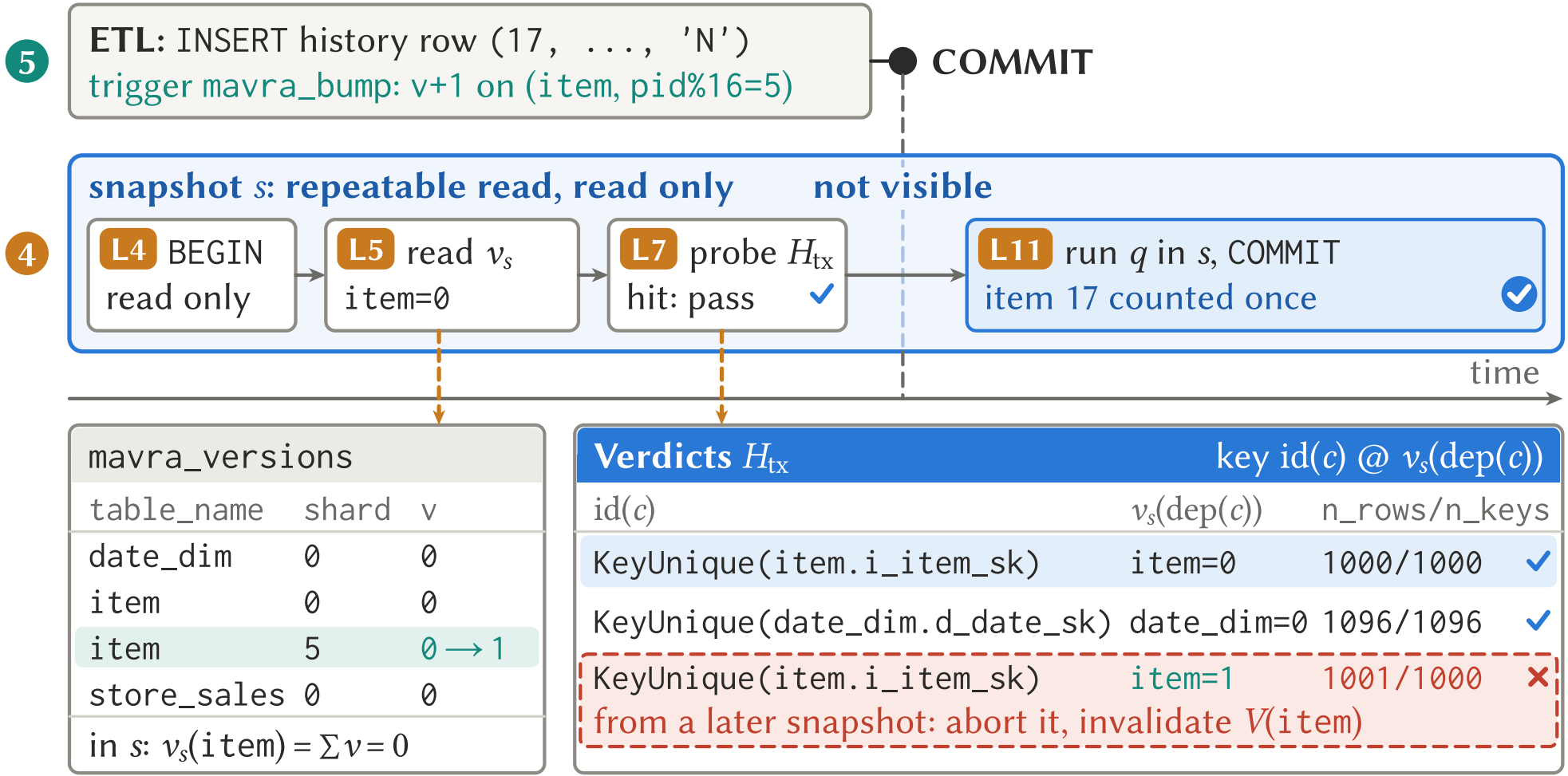

5
ETL: INSERT history row (17, ..., 'N')
trigger mavra_bump: v+1 on (item, pid%16=5)
COMMIT
snapshot s: repeatable read, read only
not visible
4
L4 BEGIN
read only
L5 read v_s
item=0
L7 probe H_{tx}
hit: pass
L11 run q in s, COMMIT
item 17 counted once
time
mavra_versions
table_name shard v
date_dim 0 0
item 0 0
item 5 0 → 1
store_sales 0 0
in s: v_s(item) = Σv = 0
Verdicts H_{tx}
key id(c) @ v_s(dep(c))
id(c)
v_s(dep(c))
n_rows/n_keys
KeyUnique(item.i_item_sk) item=0 1000/1000
KeyUnique(date_dim.d_date_sk) date_dim=0 1096/1096
KeyUnique(item.i_item_sk) item=1 1001/1000
from a later snapshot: abort it, invalidate V(item)